AI Tarot Card Reader

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1 Introduction

Artificial Intelligence (AI) has increasingly found its way into creative and recreational domains, including storytelling, music generation, and even divination tools like Tarot readings. Tarot cards have been used for centuries as a method for reflecting on life, love, and career paths [4].

In this project, we implemented an AI-powered Tarot card reader, combining classic symbolic interpretations of Tarot cards with AI techniques to create a system that simulates intuitive card selection and generates personalized reading summaries. The main problem we addressed was how to emulate the experience of a human Tarot reader using AI. Our goal was not to predict the future, but to simulate the reflective and narrative aspect of a Tarot session using interpretable, heuristic-driven methods. This includes both the symbolic selection of cards based on a chosen life category (love, life, or career) and the crafting of an insightful reading summary. The project is relevant to AI education, as it introduces simulated annealing and heuristic optimization in a creative, human-facing context.

2 Methods

2.1 Card Selection via Simulated Annealing

To simulate a Tarot reader's intuition, we used simulated annealing to generate a spread of three cards (Past, Present, Future) optimized for thematic coherence. A heuristic function scores potential spreads based on category relevance, symbolic diversity, unique keywords, and avoidance of repetition. The algorithm explores randomized combinations and gradually accepts better solutions while reducing randomness, mimicking the intuitive refinement of a human reader. This approach follows the principles outlined in standard AI literature, such as Russell and Norvig's work on stochastic optimization methods [1].

Formally, the probability of accepting a worse state is defined by:

$$P(E, T) = e^{-\frac{E}{T}} \tag{1}$$

where E is the difference in score between the current and proposed spread, and T is the current temperature. At higher temperatures, the algorithm is more likely to accept worse solutions, allowing broader exploration of the solution space.

The temperature decreases over time according to the cooling schedule:

$$T_{n+1} = \alpha \cdot T_n, \quad 0 < \alpha < 1 \tag{2}$$

where α is the cooling rate, often set between 0.85 and 0.99. This gradual reduction in T makes the algorithm increasingly selective, promoting convergence toward an optimal or near-optimal solution. This technique is commonly used in optimization problems where the solution space is large and noisy [2].The scoring function penalizes repeated card phrases, repeated themes, and overused spreads while rewarding thematic diversity and category-relevant keyword usage. This balance helps avoid generic or repetitive spreads and creates more nuanced, personalized readings. This approach follows the principles outlined in standard AI literature, such as Russell and Norvig's work on stochastic optimization methods [1].

2.2 Summary Generation

Once the spread is selected, a second simulated annealing algorithm is used to compose a meaningful summary. This process involves selecting the most relevant theme from each card, choosing one of several card meanings, and appending associated advice tailored to the selected category. Additionally, two pieces of general advice are pulled from a curated dataset. This solution is scored similarly to the spread generation and is optimized to avoid repetition while maximizing clarity, diversity, and thematic coherence.

The summary scoring function considers:

- **Category relevance**: Meanings and advice are matched to predefined keyword sets based on the chosen category (e.g., "love" or "career").
- **Theme coherence**: Summaries with varied, non-repetitive themes are rewarded.
- **Keyword alignment**: More matching keywords lead to higher scores.
- **Summary length**: Very short or overly long outputs are penalized to maintain readability.

2.3 GUI Design



Figure 1: A sample Life Reading output from the AI Tarot Card Reader, displaying three selected Tarot cards (The Fool, The Chariot, The Lovers) with their corresponding interpretations and guidance text.

The system includes a Tkinter-based graphical user interface (GUI) enhanced with sound and static card images [3] [6]. Users can select their reading category, click to reveal cards, and receive a personalized reading with visual and textual output based on predefined card meanings [5]. The GUI also supports CLI-only mode for terminal-based interactions.

The interface aims to simulate the reflective feel of a Tarot reading through features such as:

- Category buttons to choose between "Life," "Love," and "Career."
- Click-based card reveal system to display chosen cards and their meanings.
- Real-time summary generation rendered below the card layout.

Although it does not include animated card flips, the GUI presents each card one-by-one in a staged reveal, accompanied by a sound cue to simulate progression. This offers a clear and engaging experience that mirrors the pacing of an in-person Tarot session.

3 Experiments

We iteratively improved the AI selection and scoring mechanisms through targeted testing using a custom `test_ai.py` script. The script ran batch simulations and reported diversity, keyword coverage, and performance metrics.

Table 1: Improvement metrics across development stages.

Metrics	**Initial version**	**After keyword, theme**	**After spread memory, enriched summaries**
Unique Cards Pulled	15–17/22	20–21/22	22/22
Unique Spreads	10–15	20–30	77–89
Most Common Spread Freq	10–11 times	5–8 times	2–3 times
Spread Score Range	Flat (11/11/11)	Slight range (10–12)	37–45
Keyword Coverage	Inconsistent	Targeted keywords used	Strong keyword diversity
Summary Coherence	∼80% unique	∼85–87%	100% unique
Runtime Performance	Fast	Fast	Fast and consistent

Throughout development, we made several key improvements that really brought the system to life. We started by introducing a keyword-based scoring system to help the cards better match the chosen category, whether that was love, life, or career. Then we worked on reducing repetition by tweaking the scoring to avoid repeating themes or phrases too often, which made the readings feel fresher and more meaningful. We also added a memory system that kept track of previous spreads, so the same combinations wouldn't keep popping up. To make things even richer, we expanded the tarot dataset to include multiple interpretations and advice lines for each card, which helped the summaries feel more thoughtful and layered. All of these changes came together to create a system that feels smarter and more personal. The variety in card selection grew, the readings made more sense in context, and best of all, each experience felt new and different.

We also conducted informal usability tests with six users to evaluate the human experience of the AI tarot readings, using a 1–5 scale with the option to assign half-point increments.

Table 2: User Feedback on Tarot AI System (n = 6).

User	**Spread Satisfaction**	**Summary Relevance**	**Surprise Rating**
1	3.5	4.5	4
2	5	5	5
3	2.5	3.5	2.5
4	4	4.5	3.5
5	4.5	4	4
6	3	3.5	3

Average scores: Spread Satisfaction: 3.75, Summary Relevance: 4.16, Surprise Rating: 3.67

4 Results & Conclusions

Our project successfully emulated a reflective Tarot reading using simulated annealing and heuristics. The system generated meaningful, diverse, and thematically consistent card spreads and summaries. Users reported high levels of engagement, surprise, and satisfaction, especially with the summary quality. We found that symbolic diversity and category alignment were the most important factors in generating satisfying readings. Our use of general advice also added a helpful final touch that made the system feel complete.

From a technical standpoint, our results demonstrate the effectiveness of simulated annealing in low-data, symbolic environments. The modular scoring systems allowed for flexible experimentation, and the pipeline proved effective in balancing personalization and randomness. With each iteration of refinement, improvements were visible in both numerical metrics and qualitative user feedback.

4.1 Future Work

In the future, we plan to enhance the system by incorporating user feedback into the summary generation process, potentially using reinforcement learning or fine-tuned heuristics. For example, users could rate readings, and these ratings could adjust scoring weights over time.

We also intend to expand the card dataset by including multiple decks and interpretations, such as reversed card meanings and alternate symbology systems. Some portions of this project were also accelerated through ideation, code guidance, and iterative planning supported by ChatGPT [7]. Additional improvements could include:

- Integrating speech synthesis for audio-based readings using libraries like `gTTS` or `pyttsx3`.
- Enhancing the CLI interface to support export options like PDF or text logs.
- Exploring natural language generation (NLG) models for dynamic summary writing.

5 References

References

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